

BOND VALUATION, DURATION HEDGING, AND CONVEXITY-ADJUSTED RISK ANALYSIS

*Excel-Based Analysis of Term Structure Pricing and
Interest Rate Sensitivity*

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Executive Summary

Objective

This project evaluates bond valuation, interest-rate sensitivity, and hedge performance through an integrated fixed-income framework built in Excel. The analysis begins by pricing a bond using two valuation approaches: a flat-yield method and a term-structure-based method. It then extends to duration-based hedging and convexity-adjusted analysis to examine how bond values and hedge performance change under different yield shock scenarios.

The objective of the project is to study how the choice of discounting framework affects bond pricing, how bond sensitivity to interest-rate movements can be measured through duration, and how hedge effectiveness changes when second-order price effects are taken into account. By linking valuation, sensitivity, and hedging in one model, the analysis provides a practical view of how fixed-income instruments are priced and how interest-rate risk can be measured and managed.

Data & Methodology

The model was built using assumed bond inputs and market-based Treasury yield data. The bond-specific inputs included face value, coupon rate, maturity, and coupon payment frequency, which were used to define coupon cash flows and final principal repayment.

For the term-structure valuation, U.S. Treasury constant maturity yields were sourced from the Federal Reserve Economic Data (FRED) database. The selected series included DGS3MO, DGS6MO, DGS1, DGS2, DGS5, DGS10, and DGS30, which were used to construct a simplified Treasury yield curve for discounting the bond's cash flows.

Since direct observations for the 3-year and 4-year maturities were unavailable, linear interpolation was used to estimate those yields. The curve was not fully upward sloping, as short-term rates were higher than some medium-term rates, reflecting a partially inverted yield curve and a restrictive rate environment.

The analysis compared flat-yield and term-structure-based bond pricing, measured interest-rate sensitivity using Macaulay Duration, Modified Duration, and dollar duration, and then extended the framework to duration hedging and convexity-adjusted price sensitivity under different yield shock scenarios.

Assumptions

- The bond is assumed to have fixed and certain cash flows, with no default risk, embedded options, call features, or credit spread adjustments.
- Under the flat-yield method, all future cash flows are discounted using one constant yield.
- Under the term-structure method, each cash flow is discounted using maturity-specific Treasury yields.
- Since direct observations for the 3-year and 4-year maturities were unavailable, linear interpolation was used to estimate those yields.
- The interest-rate shock analysis assumes parallel shifts in yields across the curve.

- All calculations were performed in Excel so that pricing, sensitivity measures, hedge ratios, and scenario analysis could be linked in one transparent framework.

Key Analysis

The project is built around four main areas of analysis:

1. It compares flat-yield valuation with term-structure-based valuation to assess how the discounting framework affects bond price. This comparison highlights the importance of using maturity-specific rates when valuing fixed-income cash flows.
2. It examines bond sensitivity to changes in yields through Macaulay Duration, Modified Duration, and dollar duration. This establishes how changes in interest rates affect bond prices and provides the basis for measuring first-order interest-rate risk.
3. It evaluates the effectiveness of duration hedging by offsetting the dollar duration of Bond A with a position in Bond B. This allows the analysis to test whether first-order exposure can be materially reduced across a range of yield shock scenarios.
4. It extends the analysis through convexity in order to show that bond price behaviour is not purely linear. This helps explain why residual differences remain even after duration-based hedging, especially when interest-rate shocks become larger.

Overall, the project progresses from valuation to sensitivity measurement, then to hedge construction, and finally to the analysis of nonlinear bond behavior. This creates a clear fixed-income framework that connects pricing methodology with risk management outcomes.

Strategic Implications, Insights & Conclusion

The analysis highlights three main insights:

1. Bond valuation depends on the discounting framework used. While the flat-yield method provides a simple benchmark, the term-structure method offers a more refined valuation by aligning each cash flow with a maturity-specific discount rate.
2. Duration is an effective measure of first-order interest-rate risk. By matching dollar duration, the model shows that exposure to parallel yield movements can be reduced significantly, resulting in hedged profit-and-loss values that remain close to zero across multiple scenarios.
3. Convexity becomes more important when yield changes are larger. Although duration hedging reduces linear rate exposure, it does not fully capture the curvature of the bond price-yield relationship. As a result, residual differences remain even after first-order risk has been hedged.

Overall, the project shows that fixed-income analysis requires both accurate valuation and a strong understanding of interest-rate sensitivity. The combined use of term-structure pricing, duration, and convexity provides a more complete framework for evaluating bond behavior and hedge effectiveness.

Bond Valuation

Bond valuation is a core part of fixed-income analysis because the value of a bond depends on the present value of its future cash flows. These cash flows include periodic coupon payments and the repayment of principal at maturity. Since these payments occur at different points in time, the bond price depends on the discount rates used to bring those future cash flows back to present value.

In this project, the bond was valued using two different approaches: the flat-yield method and the term-structure method. Both methods are based on the same set of contractual cash flows, but they differ in the way discount rates are applied. The purpose of this comparison is to examine how the choice of discounting framework affects the estimated bond price and to determine which method provides a more realistic valuation.

Table 1: U.S. Treasury Yield Data Sourced From FRED

Maturity	Years	Yield
3M	0.25	3.72%
6M	0.5	3.70%
1Y	1	3.66%
2Y	2	3.76%
5Y	5	3.88%
10Y	10	4.27%
30Y	30	4.88%

The Treasury yield curve used in this model was constructed using U.S. Treasury constant maturity yields sourced from FRED. Since direct observations for the 3-year and 4-year maturities were not available in the selected series, linear interpolation was used to estimate the missing points. The curve was not fully upward sloping, as short-term rates were higher than some medium-term rates, reflecting a partially inverted term structure. This is relevant in bond valuation because the shape of the curve directly affects the discounting of future cash flows.

Bond A Characteristics:

Face Value - \$1,000 | Coupon Rate 5.00% | Maturity – 5 Years | Coupon Frequency – 1

Bond Valuation under Flat-Yield Method

Table 2: Bond Valuation under the Flat-Yield Method

Years	Cash Flow	Yield	Discount Factor	Present Value
1	50.00	3.88%	0.9626	48.13
2	50.00	3.88%	0.9267	46.33
3	50.00	3.88%	0.8921	44.60
4	50.00	3.88%	0.8588	42.94
5	1,050.00	3.88%	0.8267	868.02
Bond Price				1,050.03

Under the flat-yield method, all future cash flows are discounted using one constant yield. This means that every coupon payment and the final principal repayment are valued using the same

discount rate, regardless of when the cash flow occurs. The method is simple and intuitive, which makes it useful as a benchmark valuation approach.

However, the flat-yield method assumes that one single yield can represent the discounting environment for the entire life of the bond. While this simplifies the valuation process, it does not fully reflect the fact that interest rates vary across maturities. As a result, the flat-yield approach is better interpreted as a simplified pricing method rather than a fully market-aligned valuation framework.

Bond Valuation under Term-Structure Method

Table 3: Bond Valuation under the Term-Structure Method

Years	Cash Flow	Yield	Discount Factor	Present Value
1	50.00	3.66%	0.9647	48.23
2	50.00	3.76%	0.9288	46.44
3	50.00	3.80%	0.8941	44.71
4	50.00	3.84%	0.8601	43.00
5	1,050.00	3.88%	0.8267	868.02
Bond Price				1,050.41

Under the term-structure method, each bond cash flow is discounted using a maturity-specific yield taken from the Treasury curve. Shorter-dated cash flows are discounted using short-term rates, while longer-dated cash flows are discounted using yields that correspond more closely to their maturity. This creates a more refined valuation framework because the discounting process reflects the timing of each cash flow more accurately.

Compared with the flat-yield approach, the term-structure method is more consistent with how bonds are valued in practice. Since interest rates differ across maturities, applying maturity-specific discount rates provides a more realistic estimate of present value. In this project, the term-structure method incorporates both observed Treasury yields and interpolated yields to create a complete set of discount rates for the bond's cash flows.

Valuation Comparison

Table 4: Valuation Comparison across Pricing Methods

Method	Price	Difference	% Difference
Flat Yield	1,050.03		
Yield Curve	1,050.41	0.38	0.04%

The flat-yield method produced a bond price of approximately 1050.03, while the term-structure method produced a price of approximately 1050.41. The difference between the two approaches was about 0.04%. Although small, this difference shows that bond valuation is affected by the discounting framework used.

The difference arises because the flat-yield method applies one constant discount rate to all cash flows, whereas the term-structure method uses maturity-specific rates based on the yield curve. Since the curve used in this project was partially inverted, the term-structure approach captured the discounting effect more accurately.

Overall, the term-structure method provides a more robust valuation because it reflects the shape of the yield curve and aligns discount rates with the timing of each cash flow. While the flat-yield method remains useful as a simple benchmark, the term-structure approach is more consistent with practical fixed-income valuation.

Duration & Bond Sensitivity

Once the bond has been valued, the next step is to measure how sensitive that value is to changes in interest rates. Bond prices and yields move in opposite directions. When yields rise, the present value of future cash flows falls, causing the bond price to decline. When yields fall, the present value increases, causing the bond price to rise. This inverse relationship is central to fixed-income analysis and forms the basis for measuring interest-rate risk.

The degree of price sensitivity depends on the bond's cash-flow structure, coupon rate, maturity, and discount rate. In general, bonds with longer maturities and lower coupons are more sensitive to interest-rate changes because more of their value depends on cash flows received further in the future. Duration is the standard measure used to quantify this sensitivity.

Macaulay Duration, Modified Duration & Dollar Duration

Macaulay Duration measures the weighted average time required to receive a bond's cash flows. It gives a time-based view of how the bond's value is distributed across future payments. Modified Duration translates this into a direct measure of price sensitivity by estimating the approximate percentage change in bond price for a 1% change in yield.

$$Duration_{Macaulay} = \frac{\sum_{t=1}^n t \times \frac{CF_t}{(1+y)^t}}{P}$$
$$Duration_{Modified} = \frac{Duration_{Macaulay}}{1+y}$$

where:

t = Time period, CF_t = Cash Flow at time t, y = Yield per period, P = Bond price

Dollar duration expresses this same sensitivity in monetary terms. While Modified Duration shows the percentage price impact of a yield change, dollar duration converts it into the approximate dollar change in the value of the bond position. This makes it especially useful when comparing positions and building hedges.

Calculation of Macaulay Duration & Modified Duration

Bond A Characteristics:

Face Value - \$1,000 | Coupon Rate 5.00% | Maturity – 5 Years | Coupon Frequency – 1

Table 5. Bond A Duration Calculation

Years	Cash Flow	Discount Factor	Present Value	t x PV
1	50.00	0.9626	48.13	48.13
2	50.00	0.9267	46.33	92.67
3	50.00	0.8921	44.60	133.81
4	50.00	0.8588	42.94	171.75
5	1,050.00	0.8267	868.02	4,340.10

Bond B Characteristics:

Face Value - \$1,000 | Coupon Rate 4.00% | Maturity – 10 Years | Coupon Frequency – 1

Table 6. Bond B Duration Calculation

Years	Cash Flow	Discount Factor	Present Value	t x PV
1	40.00	0.9590	38.36	38.36
2	40.00	0.9198	36.79	73.58
3	40.00	0.8821	35.28	105.85
4	40.00	0.8460	33.84	135.36
5	40.00	0.8113	32.45	162.27
6	40.00	0.7781	31.12	186.75
7	40.00	0.7463	29.85	208.95
8	40.00	0.7157	28.63	229.02
9	40.00	0.6864	27.46	247.10
10	1,040.00	0.6583	684.60	6,846.04

Table 7. Duration Measures for Bond A and Bond B

Bond	Price	Macaulay Duration	Modified Duration
A	1,050.03	4.56	4.39
B	978.39	8.42	8.07

The duration measures show that Bond B is more sensitive to changes in interest rates than Bond A. This means that for the same yield change, Bond B is expected to experience a larger price movement, which makes it a suitable hedging instrument for offsetting the interest-rate exposure of Bond A.

For Bond A, a Macaulay Duration of 4.56 years means that the weighted average time to receive its cash flows is approximately 4.56 years. A Modified Duration of 4.39 means that for a 1% change in yield, the bond price is expected to change by approximately 4.39% in the opposite direction. Similarly, the higher duration measures for Bond B indicate that its value is more responsive to interest-rate changes.

These measures form the basis for the hedging framework in the next section. Once the first-order interest-rate sensitivity of each bond is quantified through duration and dollar duration, the hedge ratio can be determined by offsetting the exposure of Bond A with a position in Bond B.

Overall, duration provides a practical framework for measuring first-order interest-rate risk. By using Macaulay Duration, Modified Duration, and dollar duration together, the model captures both the timing of cash flows and the magnitude of price sensitivity.

Duration Hedging

Duration hedging is used to reduce the impact of interest-rate movements on the value of a bond position. In this project, Bond A represents the main position, while Bond B is used as the hedging instrument. The objective is to offset the first-order interest-rate risk of Bond A so that the combined portfolio becomes less sensitive to changes in yields.

The need for hedging arises because a long bond position is exposed to losses when yields rise and gains when yields fall. Since Bond A carries positive duration exposure, changes in interest rates can materially affect its value. By taking an offsetting position in Bond B, the portfolio's overall sensitivity to yield movements can be reduced.

The hedge is constructed using dollar duration, which measures the approximate dollar change in a bond's value for a 1% change in yield. While Modified Duration captures percentage price sensitivity, dollar duration expresses that same exposure in monetary terms, making it more useful for hedge design. The hedge ratio is determined by matching the dollar duration of Bond A with an offsetting position in Bond B so that the net first-order interest-rate exposure of the portfolio is brought close to zero.

$$(MV_A \times D_A) + (MV_B \times D_B) = 0$$

Where:

MV_A = Market Value of Bond A

MV_B = Market Value of Bond B

D_A = Modified Duration of Bond A

D_B = Modified Duration of Bond B

Since Bond A is the long position, Bond B must be shorted in the required amount to offset Bond A's positive duration exposure. Because Bond B has a higher modified duration than Bond A, a smaller market value of Bond B is sufficient to hedge the larger position in Bond A.

Table 8. Hedge Position Summary

Long Position in Bond A	10,00,000.00
Modified Duration of Bond A	4.39
Modified Duration of Bond B	8.07
Required Short Position in Bond B	(5,43,990.00)
Bond B Price	978.39
Units of Bond B to Short	556.00

The hedge summary shows the position required to offset the first-order interest-rate exposure of Bond A. The negative hedge value indicates that Bond B is shorted to counterbalance the positive duration exposure of the long bond position.

To evaluate hedge effectiveness, the portfolio is tested under a range of parallel yield shocks. The table below compares the profit and loss of the unhedged Bond A position with the profit and loss of the duration-hedged portfolio. This allows the hedge performance to be assessed across different interest-rate scenarios.

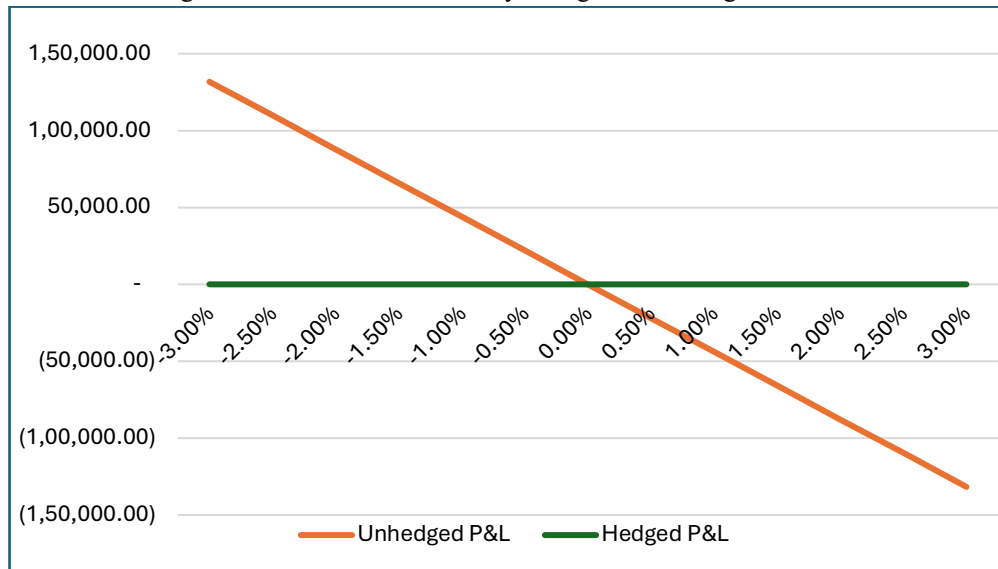
Table 9. Interest Rate Shock Analysis under Duration Hedging

Scenario	Yield Change	Bond A P&L	Bond B P&L	Unhedged P&L	Hedged P&L
-3.00%	(0.0300)	1,31,700.00	(1,31,699.98)	1,31,700.00	0.02
-2.50%	(0.0250)	1,09,750.00	(1,09,749.98)	1,09,750.00	0.02
-2.00%	(0.0200)	87,800.00	(87,799.99)	87,800.00	0.01
-1.50%	(0.0150)	65,850.00	(65,849.99)	65,850.00	0.01
-1.00%	(0.0100)	43,900.00	(43,899.99)	43,900.00	0.01
-0.50%	(0.0050)	21,950.00	(21,950.00)	21,950.00	0.00
0.00%	-	-	-	-	-
0.50%	0.0050	(21,950.00)	21,950.00	(21,950.00)	(0.00)
1.00%	0.0100	(43,900.00)	43,899.99	(43,900.00)	(0.01)
1.50%	0.0150	(65,850.00)	65,849.99	(65,850.00)	(0.01)
2.00%	0.0200	(87,800.00)	87,799.99	(87,800.00)	(0.01)
2.50%	0.0250	(1,09,750.00)	1,09,749.98	(1,09,750.00)	(0.02)
3.00%	0.0300	(1,31,700.00)	1,31,699.98	(1,31,700.00)	(0.02)

$$\text{Bond A P\&L} \approx (-D_{\text{Modified A}} \times \Delta y) \times MV_A$$

$$\text{Bond B P\&L} \approx (-D_{\text{Modified B}} \times \Delta y) \times MV_B$$

Figure 9: Interest Rate Sensitivity: Hedged vs Unhedged Portfolio



The results show that the duration hedge is highly effective in reducing the portfolio's first-order exposure to changes in interest rates. While the unhedged position experiences significant gains and losses as yields change, the hedged portfolio remains close to zero across the tested scenarios. This indicates that matching dollar duration is an effective way to neutralize linear interest-rate sensitivity.

However, the hedge does not eliminate residual differences completely. Small deviations remain, especially at larger yield shocks. This occurs because duration is a linear approximation and does not fully capture the curvature of the bond price-yield relationship. As yield movements become larger, this limitation becomes more visible.

Overall, the duration-based hedge substantially reduces the interest-rate risk of Bond A by offsetting its first-order price sensitivity with an opposing position in Bond B. This demonstrates the practical usefulness of dollar duration in hedge construction, while also highlighting the limitations of a purely linear hedging approach.

Convexity and Convexity-Adjusted Hedging

Duration is a useful measure of first-order interest-rate sensitivity, but it does not fully capture how bond prices behave when yield changes become larger. The reason is that the relationship between bond prices and yields is not linear. Instead, bond prices move along a curved path as yields change. This curvature is captured by convexity.

Convexity measures the second-order sensitivity of a bond's price to changes in yield. While duration estimates the slope of the price-yield relationship, convexity measures how that slope changes as yields move. For this reason, duration is commonly described as the first-order effect, while convexity represents the second-order effect.

The importance of convexity becomes more visible when interest-rate shocks are large. For small yield changes, duration usually provides a close approximation of bond price movement. However, as the size of the yield change increases, the limitation of the linear duration approximation becomes more apparent. Convexity helps improve this estimate by incorporating the curvature of the bond price-yield relationship.

Table 10. Convexity Measures for Bond A and Bond B

Bonds	Price	Maculay Duration	Modified Duration	Convexity
A	1,050.03	4.56	4.39	24.54
B	978.39	8.42	8.07	80.08

The convexity measures show that Bond B has higher convexity than Bond A. This means Bond B's price-yield relationship is more curved and its value responds less linearly to changes in interest rates. In practical terms, bonds with higher convexity tend to show larger deviations from duration-based estimates when yield movements become larger.

Key Formula Used:

$$\% \Delta P \approx (-D_{Modified} \times \Delta y) + \frac{1}{2} \times Convexity \times (\Delta y)^2$$
$$P\&L \approx \% \Delta P \times Market Value$$

This formula extends the duration based estimate by incorporating the second-order effect of convexity into bond price sensitivity.

Convexity-Adjusted Hedging Results

To extend the hedging analysis, convexity was incorporated into the bond price sensitivity framework. This allows the estimated price change to reflect both the first-order effect of duration and the second-order effect of convexity. The purpose of this adjustment is to examine how the hedged portfolio behaves when the nonlinear nature of bond pricing is taken into account.

In this model, the same hedge structure was retained, but the profit and loss estimates were recalculated using the convexity-adjusted price change formula. This makes it possible to compare the results of linear duration hedging with a pricing framework that explicitly recognizes curvature in the price-yield relationship.

Table 11: Interest Rate Shock Analysis under Convexity-Adjusted Hedging

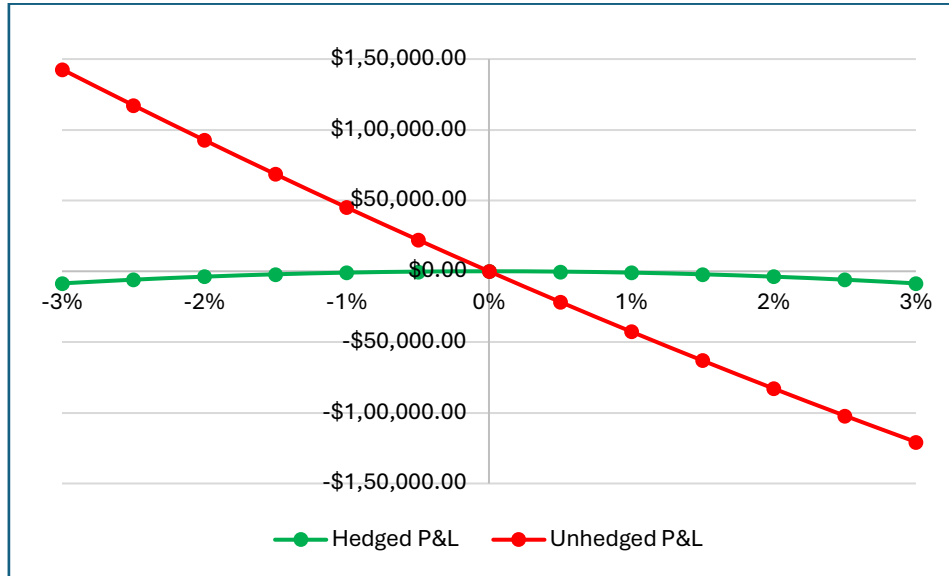
Scenario	Δy	Δy^2	% $\Delta P(A)$	P&L (A)	% $\Delta P(B)$	P&L (B)	Hedged P&L	Unhedged P&L
-3.00%	(0.0300)	0.0009	0.1427	1,42,744	0.2781	(1,51,304)	(8,560)	1,42,744
-2.50%	(0.0250)	0.0006	0.1174	1,17,420	0.2268	(1,23,364)	(5,945)	1,17,420
-2.00%	(0.0200)	0.0004	0.0927	92,708	0.1774	(96,513)	(3,805)	92,708
-1.50%	(0.0150)	0.0002	0.0686	68,611	0.1301	(70,751)	(2,140)	68,611
-1.00%	(0.0100)	0.0001	0.0451	45,127	0.0847	(46,078)	(951)	45,127
-0.50%	(0.0050)	0.0000	0.0223	22,257	0.0414	(22,495)	(238)	22,257
0.00%	-	-	-	-	-	-	-	-
0.50%	0.0050	0.0000	(0.0216)	(21,643)	(0.0393)	21,405	(238)	(21,643)
1.00%	0.0100	0.0001	(0.0427)	(42,673)	(0.0767)	41,722	(951)	(42,673)
1.50%	0.0150	0.0002	(0.0631)	(63,089)	(0.1120)	60,949	(2,140)	(63,089)
2.00%	0.0200	0.0004	(0.0829)	(82,892)	(0.1454)	79,087	(3,805)	(82,892)
2.50%	0.0250	0.0006	(0.1021)	(1,02,080)	(0.1767)	96,136	(5,945)	(1,02,080)
3.00%	0.0300	0.0009	(0.1207)	(1,20,656)	(0.2061)	1,12,096	(8,560)	(1,20,656)

The convexity-adjusted results show that the hedged portfolio continues to reduce a large portion of the unhedged interest-rate exposure, but the residual differences are more visible than in the duration-only model. This is an important result because it shows that adding convexity to the price sensitivity estimate does not automatically eliminate the remaining hedge mismatch.

The larger residuals arise because the hedge ratio was still determined using duration matching with only one hedging bond. While this neutralizes first-order interest-rate exposure, it does not fully offset second-order curvature effects. Since Bond A and Bond B have different convexity values, the portfolio remains exposed to convexity mismatch even after duration has been hedged.

This means that the convexity-adjusted model is not weaker than the duration model. Instead, it reveals the limitation of the hedge more clearly. Once the nonlinear behaviour of bond prices is included in the analysis, the remaining mismatch between the two bonds becomes easier to observe, especially under larger yield shocks.

Figure 2: Hedged vs Unhedged Portfolio under Convexity-Adjusted Sensitivity



Overall, the convexity-adjusted analysis provides a more complete view of bond price behaviour under changing interest rates. It shows that duration hedging is effective in reducing first-order exposure, but that residual risk remains when second-order effects are introduced. This highlights the importance of convexity in fixed-income risk analysis, particularly when yield movements are large or when the hedging instruments have materially different cash-flow profiles.

Comparison of Hedged P&L under Duration and Convexity-Adjusted Models

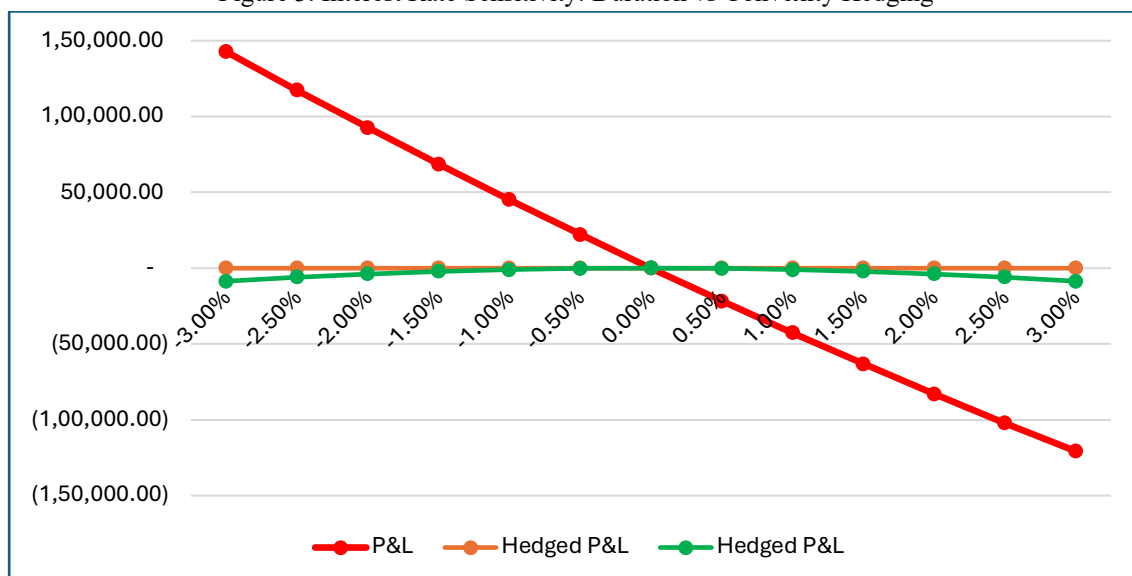
This section compares the hedged portfolio performance under the duration-based model and the convexity-adjusted model. Both approaches are designed to reduce the interest-rate exposure of Bond A by using Bond B as the hedging instrument, but they differ in how bond price sensitivity is estimated. The duration-based model captures only the first-order effect of yield changes, while the convexity-adjusted model incorporates both first-order and second-order effects.

Table 12. Comparison of Hedged P&L under Alternative Hedging Models

Yield Change	Unhedged P&L	Duration Hedged P&L	Convexity Hedged P&L
-3.00%	1,42,744.11	0.02	(8,560.29)
-2.50%	1,17,419.52	0.02	(5,944.64)
-2.00%	92,708.49	0.01	(3,804.57)
-1.50%	68,611.03	0.01	(2,140.07)
-1.00%	45,127.12	0.01	(951.14)
-0.50%	22,256.78	0.00	(237.78)
0.00%	-	-	-
0.50%	(21,643.22)	(0.00)	(237.79)
1.00%	(42,672.88)	(0.01)	(951.15)
1.50%	(63,088.97)	(0.01)	(2,140.09)
2.00%	(82,891.51)	(0.01)	(3,804.60)
2.50%	(1,02,080.48)	(0.02)	(5,944.68)
3.00%	(1,20,655.89)	(0.02)	(8,560.33)

The comparison shows that both models reduce the portfolio's exposure relative to the unhedged position, but the duration-based results remain closer to zero because the hedge was constructed by matching dollar duration. Once convexity is incorporated into the price sensitivity estimate, the remaining mismatch between the two bonds becomes more visible, especially at larger yield shocks.

Figure 3. Interest Rate Sensitivity: Duration vs Convexity Hedging



The chart shows that the duration-based hedge remains nearly flat across the tested yield shock scenarios, indicating that it is highly effective in neutralizing first-order interest-rate sensitivity. In contrast, the convexity-adjusted results show greater deviation from zero as yield changes become larger, reflecting the curvature in the bond price-yield relationship.

This does not mean that the convexity-adjusted framework is weaker. Instead, it provides a more complete view of bond behaviour by revealing the nonlinear exposure that remains after duration matching. Since the hedge ratio was based on duration alone and only one hedging bond was used, the portfolio remained exposed to convexity mismatch.

Conclusion

Bond valuation and interest-rate risk are closely connected in fixed-income analysis. This report examined that relationship by comparing flat-yield and term-structure-based pricing, measuring bond sensitivity through duration, constructing a duration hedge, and extending the analysis through convexity-adjusted price sensitivity. Together, these steps provide a practical framework for understanding how bonds are priced and how their exposure to yield movements can be evaluated.

One of the main findings is that bond valuation depends on the discounting framework used. The flat-yield method provides a simple benchmark, while the term-structure method offers a more market-aligned valuation by discounting each cash flow using a maturity-specific rate. Although the pricing difference between the two methods was small in this case, the term-structure approach remains more robust because it reflects the shape of the yield curve more accurately.

The analysis also shows that duration is an effective measure of first-order interest-rate sensitivity. By using Macaulay Duration, Modified Duration, and dollar duration, the model captures both the timing of bond cash flows and the magnitude of price sensitivity. The duration-based hedge significantly reduced the exposure of Bond A by offsetting it with an opposing position in Bond B, resulting in hedged profit and loss values that remained close to zero across the tested yield shock scenarios.

At the same time, the results show that duration alone is not sufficient when yield changes become larger. Small residual differences remained even after duration matching because duration is a linear approximation and does not fully capture the curvature of the bond price-yield relationship. Once convexity was incorporated, these second-order effects became more visible, showing that a duration-matched hedge does not fully eliminate nonlinear exposure when the two bonds have different convexity profiles.

Overall, the report highlights that fixed-income analysis requires more than bond pricing in isolation. A more complete understanding comes from linking valuation, duration, and convexity within one framework. Taken together, the results show how discounting methodology affects valuation, how first-order interest-rate exposure can be hedged, and why second-order effects remain relevant under larger yield movements.